

Subject Description Form

Subject Code	ABCT3103
Subject Title	Environmental Science
Credit Value	3
Level	3
Pre-requisite	Nil
Co-requisite	Nil
Exclusion	Nil
Objectives	This subject explains the physical and chemical processes in environments and their interactions with biological systems, including humans. These interactions constitute different types of ecosystems and ecosystem services, such as provision of clean water and nutritious food which are vital for our survival.
Intended Learning Outcomes	Upon completion of this subject, students will be able to: - realize environmental science as a multidisciplinary subject including chemistry, biology, food science and other disciplines; - understand the physicochemical and biological processes in the environment; - define ecology and ecosystems, which comprise interactions among species and the environments in which they live; - develop judgmental and analytical skills in ecological study and environmental monitoring; - understand negative human impacts, including climate change and pollution on ecosystems and biodiversity, and evaluate mitigation strategies; - appreciate the importance of environmental sustainability and the positive environmental impacts brought by humans.
Subject Synopsis/ Indicative Syllabus	<u>natural environments</u>, and their physical and chemical processes such as cycling of water and nutrients; <u>ecological processes and evolution</u>, which describe the ecological diversity, speciation/extinction, interactions among species at different levels of the ecological hierarchy and their adaptation to the environments; <u>major ecosystems</u>, ranging from macrosystems (terrestrial, aquatic and extreme environments) to microbial systems; <u>adverse environmental impacts caused by humans</u>, such as global climate change and ocean acidification, along with pollution of heavy metals, persistent organic pollutants and microplastics; and <u>environmental conservation and sustainability</u>, which explains the principles and methods in environmental monitoring, renewable energy options, novel food production in the face of environmental changes, and ecological restoration by humans as an example of positive environmental impacts.

Teaching/Learning Methodology	Interactive lectures will facilitate students’ learning of key concepts of environmental science and ecology, and promote communication between teachers and students. Tutorials will include small group discussion of relevant topics to enhance knowledge sharing among students and their awareness of recent environmental and ecological issues. Group projects will encourage students to work as a team, to research into interesting environmental and ecological topics, and to present their findings in class, and prepare written report. Assignments will be arranged to support students’ flipped classroom learning. If possible, a virtual or real field trip will be organised for students to experience the ecology of Hong Kong.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Assignment	30%					✓	✓
	2. Test	30%	✓				✓	✓
	3. Examination	40%	✓	✓	✓	✓	✓	✓
	Total	100 %						
	<u>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</u> The assignment aims to foster students’ interests in ecology and environmental conservation. Through information searching and discussion on selected topics, students will gain better insights into real-world environmental and ecological issues. The test and examination will evaluate students’ level of understanding of the taught topics. High-order thinking questions will be used to assess students’ analytical and problem-solving skills.							
Student Study Effort Expected	Class contact:							
	▪ Lecture					27 hours		
	▪ Tutorial					12 hours		
	Other study effort:							
▪ Assignment, test and self-study (and a field trip, if any)					81 hours			
Total study effort:					120 hours			
Reading List and References	1. Smith TM and Smith RL (2015) <i>Elements of Ecology</i> , 9 th ed. Pearson. QH541.S62 2015 2. Botkin DB and Keller EA (2014) <i>Environmental Science: Earth as a Living Planet</i> , 9 th ed. Wiley. GE105.B68 2014 3. Stiling PD (2002) <i>Ecology: theories and applications</i> , 4 th ed. Prentice Hall. QH541 .S675 2002							